

Progress Test 1.B — Thermo-S1

1.B · Postulates of thermodynamics

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Temperature conversion

(1 mark)

Liquid nitrogen boils at $T = 77\text{ K}$ under atmospheric pressure. Express this temperature in degrees Celsius, exactly.

[illegible]

Marking

1	
0	1

Exercise 2. Thermodynamic equilibrium — postulates

(5 marks)

A system is in contact with surroundings at pressure p_0 and temperature T_0 . Give the two conditions of thermodynamic equilibrium, with the corresponding equalities. State the postulates of classical thermodynamics. When are the state variables of a system defined?

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Equilibrium or not?

(4 marks)

A syringe with a free, sealed piston is left for a long time in a room at (p_0, T_0) . The enclosed gas then has:

- | | |
|--|--|
| ☐ $p = p_0$ and $T = T_0$ | ☐ $p > p_0$ and $T = T_0$ |
| ☐ $p = p_0$ and $T > T_0$ | ☐ p and T not defined |

Just after a sudden push on the piston, the pressure of the gas:

- ☐ already has its final value
 - ☐ is equal to p_0
 - ☐ is not defined: it is not uniform
 - ☐ is zero

A metal rod has one end in boiling water and the other in melting ice. Its temperature profile no longer changes. The rod is:

- ☐ at thermodynamic equilibrium
 - ☐ isolated
 - ☐ in a steady state, not at equilibrium
 - ☐ at equilibrium: nothing changes

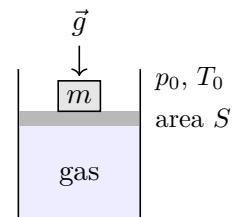
To describe a closed system at equilibrium, classical thermodynamics uses:

- ☐ all positions and velocities
 - ☐ only its mass
 - ☐ a few state variables (p , V , T)
 - ☐ its whole past history

Exercise 4. A loaded piston

(4 marks)

A vertical cylinder is closed by a frictionless piston of area S and negligible mass. The surroundings are at p_0 and T_0 . A mass m is placed on the piston. Find the pressure and the temperature of the gas once equilibrium is reached.



$$p_0 = 1.0 \times 10^5 \text{ Pa} \quad T_0 = 293 \text{ K} \quad m = 1.0 \text{ kg}$$

$$S = 2.0 \text{ cm}^2 \quad g = 10 \text{ m s}^{-2}$$

[illegible]

Marking

1

01

2

01

3

01

4

01

— End of the test —